Hartshorne Chapter 2 Section 2: Schemes

Zih-Yu Hsieh

February 23, 2026

Table of Content:

The goal for this section is to understand the notion of schemes - a generalization of the algebraic sets that was used to be defined in a more embedding sense, using the notion of sheaves, rings, and local rings.

1 Prime Spectrum and its Structure

Before starting, let's review some notion on the spectrum of a commutative ring:

Definition: Spectrum of a Commutative Ring

Given a commutative ring A, its prime spectrum $\operatorname{Spec}(A)$ is defined to be all the prime ideals $P \subseteq A$. Similarly, a maximal spectrum $\operatorname{MaxSpec}(A)$ is defined to be all the maximal ideals $M \subseteq A$.

Definition: „Closed" Sets in Spectrum

Given any ideal $I \subseteq A$, define the subset $V(I) \subseteq \operatorname{Spec}(A)$ as all the prime ideals containing I.

Which, similar to the topology on traditional algebraic sets, one can define similar notion of „topology" on $\operatorname{Spec}(A)$:

Lemma: Closed Set Relation on Spectrum

Given a commutative ring A, and $\operatorname{Spec}(A)$ the prime spectrum.
1. If $I, J \subseteq A$ are two ideals, then $V(IJ) = V(I \cap J) = V(I) \cup V(J)$.
2. If $\{I_i\}$ is a family of ideals of A, then $V(\sum I_i) = \bigcap V(I_i)$.
3. If I, J are two ideals, then $V(I) \subseteq V(J)$ iff $\sqrt{I} \supseteq \sqrt{J}$.

Proof:
1. First, for any prime ideal $P \in V(I) \cup V(J)$, since either $P \in V(I)$ or $P \in V(J)$, one has $I \subseteq P$ or $J \subseteq P$. As a result, $I \cap J \subseteq P$, hence $P \in V(I \cap J)$. This proves that $V(I) \cup V(J) \subseteq V(I \cap J)$.
   Then, since $IJ \subseteq I \cap J$, then any prime ideal $P \in V(I \cap J)$ implies $IJ \subseteq I \cap J \subseteq P$,so $P \in V(IJ)$, showing $V(I \cap J) \subseteq V(IJ)$.
   Finally, given any $P \in V(IJ)$, one must have $P \in V(I) \cup V(J)$, or else if $P \notin (V(I) \cup V(J))$, one has $P \notin V(I), V(J)$, so $I, J \not\subseteq P$. This shows there exists $i \in I \setminus P$ and $j \in J \setminus P$, which $ij \in IJ$, yet $ij \notin P$, so $IJ \not\subseteq P$ (which it's a contradiction since we assume $P \in V(IJ)$). So, $P \in V(I) \cup V(J)$, showing $V(IJ) \subseteq V(I) \cup V(J)$.

2. Given any prime ideal P, one has $P \in V(\sum I_i)$ iff $\sum I_i \subseteq P$,which happens iff $I_i \subseteq P$ for all index i, or $P \in V(I_i)$ for any index i, and finally this happens iff $P \in \bigcap V(I_i)$. Tis proves that $V(\sum I_i) = \bigcap V(I_i)$.

3. Notice that $V(I) \subseteq V(J)$ iff all prime ideals $I \subseteq P$, one has $J \subseteq P$ also. Which, it's equivalent to say any $P \in V(I)$ satisfies $J \subseteq \sqrt{J} \subseteq P$, showing that $\sqrt{J} \subseteq \bigcap_{P \in V(I)} P = \sqrt{I}$ (recall that a radical of an ideal is the intersection of all prime ideals containing it). Which, the converse holds based on the same equality.

□

Also, notice that $V(A) = \emptyset$ (since no prime ideal contains the whole ring A), and $V(0) = \mathrm{Spec}(A)$ (since all prime ideal contains 0) s a result, one can define the following topology on the spectrum:

Definition: Zariski Topology on Spectrum

Since the „algebraic set" $V(I)$ for any ideal $I \subseteq A$ satisfies the closed set relation of topology, it defines a topology on $\mathrm{Spec}(A)$, which is called **Zariski Topology**.

As a side note, for any nonzero element $f \in A$, the set $D(f) \coloneqq \{P \in \mathrm{Spec}(A) \mid f \notin P\}$ (all prime ideals not containing f) is the complement of $V((f))$, hence open; moreover it forms a basis of the Zariski Topology on $\mathrm{Spec}(A)$ (because $D(1)$ clearly covers the whole space, and $D(f) \cap D(g) = D(fg)$; finally, if $P \in U$ where $U = \mathrm{Spec}(A) \setminus V(I)$ for some ideal I, then there exists $a \in I \setminus P$, so $P \in D(a)$, while all $Q \in D(a)$ satisfies $a \notin Q$, or $I \nsubseteq Q$, hence $Q \in \mathrm{Spec}(A) \setminus V(I)$, or $P \in D(a) \subseteq U$).

Now, for the purpose of studying „morphisms" allowed on the more generalized spaces, one associates it with a sheaf of rings.

Definition: Structure Sheaf of a Spectrum

Given a commutative ring A with $X = \mathrm{Spec}(A)$. Define the **Structure Sheaf** of X, or **Sheaf of Rings** of X, as $\mathcal{O}_X \in \mathsf{Sh}(X)$, which satisfies the following:

$\mathcal{O}_X(U) = \left\{\text{functions } s : U \to \bigsqcup_{P \in U} A_P\right\}$, such that:

1. All prime ideal $P \in U$ satisfies $s(P) \in A_P$.
2. For each prime ideal $P \in U$, there exists an open neighborhood $P \in V \subseteq U$, such that there exists elements $a, f \in A$ (with $f \notin Q$ for all prime ideal $Q \in V$), satisfying $s(Q) = \frac{a}{f} \in A_Q$ for all prime ideal $Q \in V$.

Also, the restriction map is the canonical restriction as set map.

As a clarification, since U is an open subset, then every $P \in U$ there exists $f \in A$ such that $P \in D(f) \subseteq U$ (by the basis property). Which, here V can be chosen as a subset of $D(f)$ for $f \in A$ to be in the denominator.

Of course, there are things we need to check if such definition is valid as a sheaf:

Proposition

The Structure Sheaf of $\mathrm{Spec}(A)$ is a sheaf.

Proof: First, it's a presheaf of commutative rings because the functions can perform pointwise addition / multiplication (because all $s, t \in \mathcal{O}_X(U)$ with $P \in U$ satisfies $s(P), t(P) \in A_P$, hence define $(s+t)(P) \coloneqq s(P) + t(P)$, and $(s \cdot t)(P) \coloneqq s(P) \cdot t(P)$ defines a ring structure).

To verify it's a sheaf, consider the following:

- If $U \subseteq X$ is an open subset with open cover $\{V_i\}$ in U, then if $s \in \mathcal{O}_X(U)$ satisfies $s|_{V_i} = 0$ for all index i, then all prime ideals $P \in U$ (since it lies in some V_i), then $s(P) = s|_{V_i}(P) = 0$, showing that $s = 0$. This proves the separation axiom.

- Also, if every cover element has some $s_i \in \mathcal{O}_X(V_i)$ such that they agree on arbitrary finite intersection, clearly they glue to a function $s : U \to \bigsqcup_{P\in U} A_P$, where each $s(P) \in A_P$ (since if $P \in V_i$, then $s(P) = s|_{V_i}(P) = s_i(P) \in A_P$); on the other hand, for each $P \in V_i \subseteq U$, one can take the open neighborhood $P \in W \subseteq V_i$ together with $a, f \in A$ that satisfies $s_i(Q) = \frac{a}{f} \in A_Q$ for any $Q \in W$, since $s|_{V_i} = s_i$,then $s|_W = s_i|_W$, this proves the second property required for the structure sheaf. Hence, $s \in \mathcal{O}_X(U)$, showing the gluing propety.

As a result, $\mathcal{O}_X$ is indeed a sheaf of rings. □

It's kind of weird to call a presheaf a sheaf before proving it is one...anyway, after this we can observe some nice property of this structure sheaf:

Theorem

Given A a commutative ring, $(X = \mathrm{Spec}(A), \mathcal{O}_X)$ the spectrum of A with its structure sheaf.

1. For any prime ideal $P \in \mathrm{Spec}(A)$, the stalk $\mathcal{O}_{X,P}$ is isomorphic to the local ring A_P.
2. For any element $f \in A$, the ring $\mathcal{O}_X(D(f))$ is isomorphic to the localized ring A_f.
3. One has $\mathcal{O}_X(A) \cong A$.

Proof:

1. Notice that if consider the direct limit $\mathcal{O}_{X,P}$, every element $s_P \in \mathcal{O}_{X,P}$ has some representative $s \in \mathcal{O}_X(U)$ for some open subset $U \subseteq X$. Which, one can define the following map $\mathrm{ev} : \mathcal{O}_{X,P} \to A_P$ by $\mathrm{ev}(s_P) := s(P) \in A_P$ (which is well-defined, as all representative can agree on some open neighborhood of P, in particular they all agree on P), and it's a ring homomorphism just by definition of additions and multiplications on each $\mathcal{O}_X(U)$.

   - To prove surjectivity, notice that for all $\frac{a}{f} \in A_P$, with $f \notin P$, one has $P \in D(f)$; as a result, choose $D(f) \subseteq X$ as the open neighborhood, then the function $s : D(f) \to \bigsqcup_{Q\in D(f)} A_Q$ by $s(Q) = \frac{a}{f} \in A_Q$ is well-defined (as $Q \in D(f)$ implies $f \notin Q$), and take $V = D(f)$, with the chosen element $a, f \in A$, it clearly satisfies the second condition for the structure sheaf.

     Hence, take its image $s_P \in \mathcal{O}_{X,P}$, one has $\mathrm{ev}(s_P) = s(P) = \frac{a}{f}$, showing surjectivity of ev.

   - To prove injectivity, suppose $\mathrm{ev}(s_P) = 0$, then its representative has $s(P) = 0$. Which, using the structure sheaf property, there exists some open neighborhood $P \in V \subseteq U$, together with $a, f \in A$ (with $f \notin Q$ for any $Q \in V$), such that $s(Q) = \frac{a}{f} \in A_Q$ for all $Q \in V$. As a result, $s(P) = \frac{a}{f} = 0 \in A_P$, which by the definition of localization, there exists $u \in A \setminus P$, such that $(a \cdot 1 - 0 \cdot f)u = 0$, or $au = 0$ (where $0 \in P$ is by definition). Using the prime property, $u \notin P$ implies that $a \in P$.

     Now, notice that this generalizes to any prime ideal in $V \cap D(f) \cap D(u)$ (since if $Q \in V \cap D(f) \cap D(u)$, one first has $s(Q) = \frac{a}{f}$ by definition of V, and one also has $f, u \notin Q$, showing that $(a \cdot 1 - 0 \cdot f)u = 0$, or $\frac{a}{f} = 0$ in A_Q). Hence, within this open neighborhood, one has s restricted to 0, showing that $s_P = 0$. This proves injectivity.

2. For this, we'll define another homomorphism $\psi : A_f \to \mathcal{O}_X(D(f))$, by mapping all $\frac{a}{f^n} \in A_f$, to a map $\psi_{\frac{a}{f^n}} : D(f) \to \bigsqcup_{P\in D(f)} A_P$ by $\psi_{\frac{a}{f^n}}(P) := \frac{a}{f^n} \in A_P$. This is a well-defined ring homomorphism, just because addition and multiplication can be performed pointwise; and, it clearly satisfies the second condition for structure sheaf (by choosing $V = D(f)$, and $a, f^n \in A$).

- To prove injectivity, suppose one has $\psi_{\frac{a}{f^n}} = 0$, then for any prime ideal $P \in D(f)$, one has $\frac{a}{f^n} = 0 \in A_P$. So, there exists $u_P \in A \setminus P$, such that $(a \cdot 1 - 0 \cdot f^n)u_P = 0$, or $au_P = 0$. If consider the ideal $J = \operatorname{Ann}(a)$, one has $u_P \in J$, while $u_P \notin P$, this shows that $J \nsubseteq P$, or $P \notin V(J)$.

  Notice that such relation is true for all $Q \in D(f)$, hence one has $V(J) \cap D(f) = \emptyset$. As a result, this implies $V(J) \subseteq V((f))$, in other words $f \in \sqrt{J}$, so there exists some power $m \in \mathbb{N}$, such that $f^m \in J = \operatorname{Ann}(a)$, or $af^m = 0$. Hence, within the localization A_f, one has $(a \cdot f^m - 0 \cdot f^n) = 0$, showing $\frac{a}{f^n} = 0$. This proves the injectivity of ψ.

- To prove surjectivity, given any map $s : D(f) \to \bigsqcup_{P \in D(f)} A_P$, by the definition of structure sheaf, each $P \in D(f)$ has an open neighborhood $P \in V_P \subseteq D(f)$ with elements $a_P, g_P \in A$, such that $s(Q) = \frac{a_P}{g_P}$ for all $Q \in V_P$ (in particular, each $g_P \notin Q$ for all $Q \in V_P$). This indicates that $V_P \subseteq D(g_P)$; which, by restricting to the basis element, WLOG one can assume $V_P = D(h_P)$ for some $h_P \in A$.

  Then, since $D(h_P) \subseteq D(g_P)$, one has $V((h_P)) \supseteq V((g_P))$, indicating that $\sqrt{(h_P)} \subseteq \sqrt{(g_P)}$. Hence, there exists $m_P \in \mathbb{N}$, such that $h_P^{m_P} \in (g_P)$, or there exists $c_P \in A$, such that $h_P^{m_P} = c_P g_P$. Which, one concludes that $\left(a_P \cdot h_P^{m_P} - a_P \cdot c_P \cdot g_P\right) = 0$, or $\frac{a_P}{g_P} = \frac{c_P a_P}{h_P^{m_P}}$. This shows that on $D(h_P)$ particularly, one has $\frac{a_P}{g_P} = \frac{c_P a_P}{h_P^{m_P}}$ for each prime ideal in $D(h_P)$.

  Since $D(h) = D(h^n)$ for any nonzero $h \in A$ and $n \in \mathbb{N}$ (using the prime property), one can replace $h_P^{m_P}$ with h_P (for notation simplicity), and $c_P a_P$ by a_P. So, one has s being represented as $\frac{a_P}{h_P}$ on the open subset $D(h_P)$.

  Next, we'll consider reducing the number of $D(h_P)$'s to be finite: Given that $D(f) = \bigcup D(h_P)$, then one has $V((f)) = \bigcap V((h_P)) = V(\sum(h_P))$. So, one has $f \in \sqrt{\sum(h_P)}$, showing that $f^n \in \sum(h_P)$ for some natural number n. This further implies that there exists finite list $h_1, ..., h_m$ within the h_P's (with coefficients $b_1, ..., b_m \in A$), such that $f^n = \sum_{i=1}^m b_i h_i$.

  This indicates that $f \in \sqrt{\sum_{i=1}^m (h_i)}$, as a result $V((f)) \supseteq V\left(\sum_{i=1}^m (h_i)\right) = \bigcap_{i=1}^m V((h_i))$. So, $D(f) \subseteq \bigcup_{i=1}^m D(h_i)$, showing finite choice suffices.

  Now, combining the previous two facts, one has $D(f)$ being covered by $D(h_1), ..., D(h_m)$, and for each of them s is represented by $\frac{a_i}{h_i}$ for some element $a_i \in A$. Then, if consider $D(h_i) \cap D(h_j) = D(h_i h_j)$ (and the restriction of s onto this), the two elements $\frac{a_i}{h_i}$ and $\frac{a_j}{h_j}$ represents the same element on $D(h_i h_j)$. Then, consider the ring homomorphism $\psi : A_{h_i h_j} \to \mathcal{O}_X\big(D(h_i h_j)\big)$, the representative of s implies that $s|_{D(h_i h_j)} = \psi_{\frac{a_i}{h_i}} = \psi_{\frac{a_j}{h_j}}$, hence with injectivity of ψ, one must have $\frac{a_i}{h_i} = \frac{a_j}{h_j}$ in $A_{h_i h_j}$. As a result, there exists $l_{ij} \in \mathbb{N}$, such that the following holds:

  $$\left(a_i \cdot h_j - a_j \cdot h_i\right)\left(h_i h_j\right)^{l_{ij}} = 0 \tag{1.1}$$

  Notice that the collection $\{h_i\}$ is finite now, so one can choose N as the maximum for all l_{ij}, deducing the following equation for arbitrary i, j:

  $$\left(a_i \cdot h_j - a_j \cdot h_i\right)\left(h_i h_j\right)^N = 0, \quad \left(a_i h_i^N\right)\left(h_j^{N+1}\right) - \left(a_j h_j^N\right)\left(h_i^{N+1}\right) = 0 \tag{1.2}$$

  Finally, replacing each a_i with $h_i^N a_i$, and h_i with h_i^{N+1} (or, denote these replacement as a_i, h_i respectively), then one deduces that each $\frac{a_i}{h_i}$ (after the replacement) still represents the original element, and now we can assume $a_i h_j = a_j h_i$ for any i, j.

  Write $f^n = \sum_{i=1}^m b_i h_i$ as before (with some newly chosen b_i), if consider $a = \sum_{i=1}^m b_i a_i$, then it deduces the following:

$$ah_j = \sum_{i=1}^{m} b_i a_i h_j = \sum_{i=1}^{m} b_i a_j h_i = a_j \sum_{i=1}^{m} b_i h_i = a_j f^m \tag{1.3}$$

So, this proves that the element $\frac{a}{f^m} = \frac{a_j}{h_j}$ on $D\big(h_j\big)$ (within the prime ideals). So, $\psi_{\frac{a}{f^n}} = s$ on the whole domain, showing the surjectivity of ψ.

This concludes the isomorphism between A_f and $\mathcal{O}_X(D(f))$.

3. For the third case, notice that $A_f = A$ for $f = 1$. Hence, choosing $D(1) = \mathrm{Spec}(A)$, one has $\mathcal{O}_X(\mathrm{Spec}(A)) \cong A_1 = A$. Which, it's a special case of 2.

□

2 Ringed Space and Locally Ringed Space

With the definitions of prime spectrum, we'd like to have a category regulating simillar objects:

Definition: Category of Ringed Space

A **Ringed Space** is a pair $(X, \mathcal{O}_X)$, where X is a topological space, and $\mathcal{O}_X$ is its structure sheaf (sheaf of ring).

A **morphism** of ringed spaces $\left(f, f^{\#}\right) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ consists of a continuous map $f : X \to Y$, and morphism of sheaves $f^{\#} : \mathcal{O}_Y \to f_* \mathcal{O}_X$ of sheaves of rings on Y.

Together, this forms a category of Ringed Spaces.

For extra constraints, we also want to restrict the stalks:

Definition: Category of Locally Ringed Space

One define a ringed space $(X, \mathcal{O}_X)$ as a **Locally Ringed Space**, if for each point $P \in X$, the stalk $\mathcal{O}_{X,P}$ is a local ring.

Also, a morphism $\left(f, f^{\#}\right) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a **morphism of Locally Ringed Space**, if the induced homomorphism $f^{\#} : \mathcal{O}_{Y,f(P)} \to f_* \mathcal{O}_{X,P}$ is a *local homomorphism* of local rings.

For clarification, first let's explain the construction of the morphisms on stalks: Given any open subset $V \subseteq Y$, one has $V \ni f(P)$ implies $f^{-1}(V) \ni P$. Hence, if consider the direct limit over a system of $V \ni f(P)$, hen one has a ring homomorphism $\mathcal{O}_{Y,f(P)} \to \lim_{V \ni f(P)} \mathcal{O}_X(f^{-1}(V))$. Which, because the directed system span over a subportion of all $f_* \mathcal{O}_X(V) = \mathcal{O}(f^{-1}(V))$ (with $f^{-1}(V) \ni P$), one generates a ring homomorphis $\lim_{V \ni f(P)} \mathcal{O}_X(f^{-1}(V)) \to f_* \mathcal{O}_{X,P}$. Which, the composition generates the desired orphism on stalks.

Also, given A, B two local rings (wht m_A, m_B the maximal ideals respectively), a ring homomorphism $f : A \to B$ is called a *local homomorphism* if $\varphi^{-1}(m_B) = m_A$ (or equivalently, one has $m_A^e \subseteq m_B$).

As a side note, within this category, $\left(f, f^{\#}\right)$ is an isomorphism (with two-sided inverse) iff f is a topological homeomorphism, with $f^{\#}$ being an isomorphism of sheaves.

Let's get some examples regarding ringed spaces and locally ringed spaces:

Example: Ringed Space that is not Local

Consider a connected topological space X, with the constant sheaf of $\mathbb{Z}$, say $F_{\mathbb{Z}}$ equipped with. Then, $(X, F_{\mathbb{Z}})$ is a ringed space, but with any localization $F_{\mathbb{Z},x} \cong \mathbb{Z}$, which is not local.

So, it's not a locally ringed space.

Example: Morphism of Ringed Space, but not Local

Given a local integral domain A (for instance, $A = \mathbb{Z}_{(p)}$), together with its fraction field $K = F(A)$, there is an obvious inclusion $\iota : A \hookrightarrow K$, and notice that as an integral domain, for any multiplicatively closed set $S \subset A$, the incuded localization map $A \to S^{-1}A$ is injective, and $S^{-1}A$ includes into K by definition.

Now, consider $X = \{*\}$ a one point space, define F_A, F_K to be the constant sheaf of A and K respectively on X, then $(X, F_K), (X, F_A)$ are naturally locally ringed spaces. Which, define the pair $(\mathrm{id}_X, f) : (X, F_K) \to (X, F_A)$, where $\mathrm{id}_X : X \to X$ is the identity as topological homeomorphism, and $f : F_A \to F_K$ is the following morphism:

$$\begin{aligned} \forall U \subseteq X \text{ open}, \quad & f(U) : F_A(U) = \mathrm{Hom}_{\mathsf{Top}}(U, A) \to F_K(U) = \mathrm{Hom}_{\mathsf{Top}}(U, K) \\ & f(U)(\psi) = \iota \circ \psi \end{aligned} \tag{2.1}$$

Notice that since A, K are endowed with discrete topology, $\iota : A \hookrightarrow K$ is naturally a continuous map, so the above map f is well-defined. So, they actually form a morphism of ringed spaces.

Yet, it's not a morphism of locally ringed spaces, as the direct limit of both sheaves are A, K respectively, so the induced map on stalk $\overline{f} : F_{A,*} \to F_{K,*}$ becomes $\overline{f} : A \to K$, and it's precisely the inclusion, which is not a local homomorphism.

Regarding the theorem of prime spectrum's structure, one has the following characterization:

Theorem: Prime Spectrum as Locally Ringed Space

1. Given A a commutative ring, then $(\mathrm{Spec}(A), \mathcal{O})$ is a locally ringed space.
2. If $\varphi : A \to B$ is a commutative ring homomorphism, then the map $\varphi' : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ by $\varphi'(P) = \varphi^{-1}(P)$ is a continuous map, and it induces a morphism of locally ringed spaces
$$\left(f, f^{\#}\right) : (\mathrm{Spec}(B), \mathcal{O}_B) \to (\mathrm{Spec}(A), \mathcal{O}_A) \tag{2.2}$$
3. If A, B are commutative rings, then any morphism of locally ringed spaces $\left(f, f^{\#}\right) : (\mathrm{Spec}(B), \mathcal{O}_B) \to (\mathrm{Spec}(A), \mathcal{O}_A)$ is induced by a ring homomorphism $\varphi : A \to B$.

Proof:

1. This follows the fact that $\mathcal{O}_P \cong A_P$ for all $P \in \mathrm{Spec}(A)$ (cf. previous theorem).

2. Given a commutative ring homomorphism $\varphi : A \to B$, notice the map $f : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is continuous: Suppose $I \subseteq A$ is an ideal, and consider the closed set $V(I) \subseteq \mathrm{Spec}(A)$. Then, one has $P \in f^{-1}(V(I))$ iff $f(P) = \varphi^{-1}(P) \in V(I)$ iff $\varphi(I) \subseteq P$. Hence, take $V(I^e)$, one has $P \in V(I^e)$ iff $\varphi^{-1}(P) \in V(I)$, showing that $f^{-1}(V(I)) = V(I^e)$, which f is a continuous map.

   Then, notice that given any prime ideal $P \in \mathrm{Spec}(B)$, the composition of localization map with φ, say $\pi \circ \varphi : A \to B \to B_P$ has all element in $A \setminus \varphi^{-1}(P)$ being invertible, so it descends to a homomorphism $\varphi_P : A_{\varphi^{-1}(P)} \to B_P$ (which is a local homomorphism, since $\varphi_P(\varphi^{-1}(P)) = \varphi(\varphi^{-1})(P)B_P \subseteq PB_P$).

   Which, for any open subset $U \subseteq \mathrm{Spec}(A)$, take $f^{-1}(U)$, construct the following map:
$$\begin{aligned} & f_U^{\#} : \mathcal{O}_A(U) \to \mathcal{O}_B(f^{-1}(U)), \quad \forall s : U \to \bigsqcup_{Q \in U} A_Q \\ & f_U^{\#}(s) : f^{-1}(U) \to \bigsqcup_{P \in f^{-1}(U)} B_P, \quad f_U^{\#}(s)(P) = \varphi_P \circ s \circ f(P) = \varphi_P \circ s(\varphi^{-1}(P)) \end{aligned} \tag{2.3}$$

   Which, it's clear that $f_U^{\#}(s)(P) \in B_P$ by the definition (since $s(\varphi^{-1}(P)) \in A_{\varphi^{-1}(P)}$, so it's in the domain of φ_P). To verify that it's a ring homomorphism, consider the following:
$$\begin{aligned} f_U^{\#}(s+t)(P) &= \varphi_P \circ (s+t)(\varphi^{-1}(P)) = \varphi_P(s(\varphi^{-1}(P)) + t(\varphi^{-1}(P))) \\ &= \varphi_P \circ s(\varphi^{-1}(P)) + \varphi_P \circ t(\varphi^{-1}(P)) = f_U^{\#}(s)(P) + f_U^{\#}(t)(P) \end{aligned} \tag{2.4}$$
$$\begin{aligned} f_U^{\#}(s \cdot t)(P) &= \varphi_P \circ (s \cdot t)(\varphi^{-1}(P)) = \varphi_P(s(\varphi^{-1}(P)) \cdot t(\varphi^{-1}(P))) \\ &= \varphi_P(s(\varphi^{-1}(P))) \cdot \varphi_P(t(\varphi^{-1}(P))) = f_U^{\#}(s)(\varphi^{-1}(P)) \cdot f_U^{\#}(t)(\varphi^{-1}(P)) \end{aligned} \tag{2.5}$$

Which, $f_U^\#(s+t) = f_U^\#(s) + f_U^\#(t)$, and $f_U^\#(s \cdot t) = f_U^\#(s) \cdot f_U^\#(t)$, showing $f_U^\#$ is a ring homomorphism.

Which, with the restriction map being restriction of domain, the diagram clearly commutes:

$$\begin{tikzcd} \mathcal{O}_A(U) \arrow[r, "f_U^\#"] \arrow[d, "\rho_{UV}"'] & \mathcal{O}_B(f^{-1}(U)) \arrow[d, "\rho_{UV}"] \\ \mathcal{O}_A(V) \arrow[r, "f_V^\#"'] & \mathcal{O}_B(f^{-1}(V)) \end{tikzcd}$$

And the reason is because any $P \in f^{-1}(V)$ has $f(P) = \varphi^{-1}(P) \in V$, which the restriction of U to V for any $s \in \mathcal{O}_A(U)$, can be carried over when restricting $f^{-1}(U)$ to $f^{-1}(V)$ for $\mathcal{O}_B(f^{-1}(U))$.

Hence, this defines a morphism of sheaf of rings $f^\# : \mathcal{O}_A \to f_*\mathcal{O}_B$.

Finally, this is a morphism of locally ringed space, because any $P \in \operatorname{Spec}(B)$ has the descended map being $\varphi_P : A_{\varphi^{-1}(P)} \to B_P$, which is already proven as a local homomorphism.

3. Now, consider an morphism of locally ringed spaces $(f, f^\#) : (\operatorname{Spec}(B), \mathcal{O}_B) \to (\operatorname{Spec}(A), \mathcal{O}_A)$, notice that take the whole space, one has a ring homomorphism $\varphi = f^\#_{\operatorname{Spec}(A)} : \mathcal{O}_A(\operatorname{Spec}(A)) \to \mathcal{O}_B(f^{-1}(\operatorname{Spec}(A))) = \mathcal{O}_B(\operatorname{Spec}(B))$, which reads precisely as $\varphi : A \to B$. Now, we claim that φ induces a morphism of locally ringed spaces (cf. part (b)) that's identical to $(f, f^\#)$.

   First, for any $P \in \operatorname{Spec}(B)$, notice the map $\varphi : A \to B$ descends to a local homomorphism $\varphi_P : A_{\varphi^{-1}(P)} \to B_P$, which is of the following commutative diagram:

$$\begin{tikzcd} A \arrow[r, "\varphi"] \arrow[d] & B \arrow[d] \\ A_{\varphi^{-1}(P)} \arrow[r, "\varphi_P"'] & B_P \end{tikzcd}$$

   On the other hand, the morphism of sheaf of rings $f^\# : \mathcal{O}_A \to f_*\mathcal{O}_B$ also has a local homomorphism $f_P^\# : \mathcal{O}_{A,f(P)} \to \mathcal{O}_{B,P}$ (or, $f_P^\# : A_{f(P)} \to B_P$), which is descended from the direct limit (where the direct limit contains open subset $\operatorname{Spec}(B) \ni P$ for B_P, and $\operatorname{Spec}(A) \ni f(P)$ for $A_{f(P)}$), hence one also has the following commutative diagram:

$$\begin{tikzcd} A \arrow[r, "\varphi"] \arrow[d] & B \arrow[d] \\ A_{f(P)} \arrow[r, "f_P^\#"'] & B_P \end{tikzcd}$$

   Now, notie that this enforces $f(P) = \varphi^{-1}(P)$ (since if $f(P) \neq \varphi^{-1}(P)$, then there exists element lying in one but not the other; for definiteness, say there exists $a \in \varphi^{-1}(P) \setminus f(P)$. Then, the image of a in B_P is not invertible, since $a \notin A \setminus \varphi^{-1}(P)$, which by the first diagram is not allowed to be inverted; on the other hand, a in B_P should be invertible, since $a \in A \setminus f(P)$, which by the second diagram is being inverted, which causes a contradiction).

   Then, this shows that $f : \operatorname{Spec}(B) \to \operatorname{Spec}(A)$ coincides with the map induced by $\varphi : A \to B$, which further follows that $f^\#$ must be induced by φ also. Therefore, $(f, f^\#)$ is induced by a ring homomorphism $A \to B$ (and the particular choice is $f^\#_{\operatorname{Spec}(A)} : A \to B$).

□

3 Affine Schemes and Schemes

With all of these, one can make a definition of the most general object in Algebraic Geometry:

Definition: Affine Schemes and Schemes

An **Affine Scheme** is a locally ringed space $(X, \mathcal{O}_X)$ that is isomorphic to $(\operatorname{Spec}(A), \mathcal{O}_A)$ for some commutative ring A (as locally ringed spaces).

A **Scheme** is a locally ringed space $(X, \mathcal{O}_X)$, such that every point has an affine neighborhood. More precisely, for every $x \in X$, it has an open neighborhood $x \in U \subseteq X$, so that the restriction of sheaves $\mathcal{O}_X \mid_U$ satisfies $(U, \mathcal{O}_X \mid_U)$ being an affine scheme.

(Equivalently, a Scheme is a locally ringed space with an affine cover).

Which, the morphism is restricted to morphism of locally ringed spaces, and an isomorphism is an isomorphism of locally ringed spaces, which s DIFFERENT from topological homeomorphism (EX: morphisms between spectrum of non-isomorphic fields).

Finally, before the end of the section, let's talk about some examples / definitions that'll be motivated:

Definition: Affine n-Space

Given k a field, define $\operatorname{Spec}(k[x_1, ..., x_n]) =: \mathbb{A}_k^n$ as the **Affine n-Space**. For $n = 1$, the space $\mathbb{A}_k^1 = \operatorname{Spec}(k[x])$ is called an **Affine Line**.

As a side note, all commutative ring's spectrum is a „Spectral Space", which also has a unique „generic point" (with a point whose closure is the whole space). In these cases, the zero ideal is the generic point.

Also, if k is algebraically closed, then $\mathbb{A}_k^1 \setminus \{0\}$ (removing the zero ideal) is classified by all maximal ideal $(x - a)$ (where $a \in k$), which taking away the generaic point, the affine line has a one-to-one corerspondance with k itself. Which creates an analogy to traditional affine 1-space (each point corresponds to a maximal ideal in $k[x]$, and the whole space corresponds to the zero ideal).

Example: Affine Plane

By definition, $\mathbb{A}_k^2 = \operatorname{Spec}(k[x, y])$. Which, the point is closed $\Longleftrightarrow$ it's maximal. If k is algebraically closed, then Hilbert's Nullstellensatz guarantees a one-to-one correspondance of closed points in $\mathbb{A}_k^2$ and all maximal ideals. Which, with the induced topology, it's homeomorphic to k^2 (under Zariski Topology), via the map $C \to k^2$ of $(x - a, y - b) \mapsto (a, b)$.

As a bijection it's clear; given any algebraic set $Z \subset k^2$, it's the zero of some polynomials $f_1, ..., f_n \in k[x, y]$, which take $V((f_1, ..., f_n))$ and intersect with C the closed points, it's precisely all maximal ideal $(x - a, y - b)$ containing $(f_1, ..., f_n)$, and as a result all $f_i(a, b) = 0$, or $(a, b) \in Z$, showing $(x - a, y - b)$ is in the preimage of Z, therefore the map is continuous.

Finally, the map is also a closed map, as if the collection of maximal ideals are precisely $V(I) \cap C$ (closed in subspace topology of C), then every polynomial $f \in I$ is contained in $(x - a, y - b)$ (if this ideal is in the intersection), as a result $f(a, b) = 0$. So, $(a, b) \in Z(I) \subset k^2$; moreover, if $(a, b) \in Z(I)$, then $(x - a), (y - b) \in \sqrt{I}$ using Nullstellensatz, showing that $V(I) = V\left(\sqrt{I}\right) \ni (x - a, y - b)$. So, the image of $V(I) \cap C$ is precisely $Z(I) \subset k^2$, generating the homeomorphism.

Also, given any irreducible polynomial $f \in k[x, y]$, since $V((f)) \cong \operatorname{Spec}(k[x, y]/(f))$ as topological space, notice that the zero ideal $0 \in \operatorname{Spec}(k[x, y])/(f)$ has the closure being the whole space, as a result the corresponding prime ideal $(f) \in V((f))$ is the generic point.

With (f) being at least height 1 (and $k[x, y]$ has height 2), then any other distinct point from (f) in $V((f))$ must be closed points (i.e. maximal ideals). So, one has $(f) \in V((f))$ be a generic point (of the subspace topology of $V((f))$), whose closure contains only (f) and other closed point (that as mximal ideal contains (f)).

(Note: Here (f) is in fact THE generic point, as $\operatorname{Spec}(k[x, y]/(f))$ only has one generic point).

Finally, to construct a nicer structure, one would like a way of „gluing“ schemes together to form new schemes. Using the categorical logic, when there are subschemes that are isomorphic, it's desirable to have fibre coproduct of them (by identifying the subschemes as the same).

For the topological purpose, let's review some terminology from topology:

Definition: Adjoint Space

In the category of Topological Spaces, given two topological space X, Y, and their open subsets $U \subseteq X$, $V \subseteq Y$ are homeomorphic (via a map $\varphi : U \xrightarrow{\sim} V$), then the adjoint space $X \sqcup_\varphi Y$ is defined as $(X \sqcup Y)/\sim$, where every $a \in U$ has $a \sim \varphi(a)$ in $X \sqcup Y$.

The topology on the adjoint space $X \sqcup_\varphi Y$ is the quotient topology of the disjoint union space $X \sqcup Y$.

Notice that this is the fibre coproduct of the following diagram:

$$\begin{tikzcd}
U \arrow[r, hook] \arrow[d, hook, "\varphi"'] & X \arrow[d, hook] \arrow[ddr, bend left, "i_X"] & \\
Y \arrow[r, hook] \arrow[drr, bend right, "i_Y"'] & X \sqcup_\varphi Y \arrow[dr, dotted, "\exists! i"] & \\
& & Z
\end{tikzcd}$$

Since if any topological space Z with continuous maps $i_X : X \to Z$ and $i_Y : Y \to Z$ that satisfies $i_X \circ \iota : U \to Z$ and $i_y \circ \varphi : U \to Z$ agrees, then any continuous map $i : X \sqcup_\varphi Y \to Z$ lifts to a continuous map $\bar{i} : X \sqcup Y \to Z$, such that the composition with inclusion $Y \hookrightarrow X \sqcup Y$ and $X \hookrightarrow X \sqcup Y$ recovers the map i_Y, i_X respectively (using the property of disjoint union space).

Which, for any $a \in X \setminus U$, one must have $i(a) = i_X(a)$; for all $b \in Y \setminus V$, one must have $i(b) = i_Y(b)$; and finally, for all $\varphi(c) \sim c \in U$ (where $U \cong V$ via φ), one has $i(c) = i_X(c) = i_Y \circ \varphi(c)$, showing the uniqueness of i as a set map.

And, the existence of i (as continuous map) is based on the fact that any open subset $W \subseteq Z$ has preimage $i_X^{-1}(W) \subseteq X$, $i_Y^{-1}(W) \subseteq Y$ being open, so $(i_X^{-1}(W) \sqcup i_Y^{-1}(W)) \subseteq (X \sqcup Y)$ is open. Hence, its quotient in $X \sqcup_\varphi Y$ is also open (which is precisely the preimage under i). This shows the continuity of i.

Now, using such idea, we can construct „gluing of schemes“:

Theorem: Gluing of Schemes

Let X, Y be two schemes (here we omit the structure sheaf for simplicity), and let $U \subseteq X$, $V \subseteq Y$ be two open subsets, such that there exists an isomorphism of schemes $(\varphi, \varphi^\#) : (U, \mathcal{O}_X|_U) \xrightarrow{\sim} (V, \mathcal{O}_Y|_V)$. Then, the adjoint space $X \sqcup_\varphi Y$ has a natural scheme, together with morphisms of schemes $(\iota_X, \iota_X^\#) : (X, \mathcal{O}_X) \to (X \sqcup_\varphi Y, \mathcal{O}_\varphi)$ and $(\iota_Y, \iota_Y^\#) : (Y, \mathcal{O}_Y) \to (X \sqcup_\varphi Y, \mathcal{O}_\varphi)$, such that the following diagram commutes:

$$\begin{tikzcd}
U \arrow[r, hook] \arrow[d, hook, "\varphi"'] & X \arrow[d, hook, "\iota_X"] \arrow[ddr, bend left, "i_X"] & \\
Y \arrow[r, hook, "\iota_Y"'] \arrow[drr, bend right, "i_Y"'] & X \sqcup_\varphi Y \arrow[dr, dotted, "\exists! i"] & \\
& & Z
\end{tikzcd}$$

Where, all the arrows are morphism of schemes.

Proof: It's clear that $X \sqcup_\varphi Y$ is a topological space. So, the only task is to construct a suitable structure sheaf with the desired property.

Given any open subset $W \subseteq X \sqcup_\varphi Y$, notice its preimage in U (i.e. $W \sec U = W \cap V$ in $X \sqcup_\varphi Y$, where U is identified as a subspace of $X \hookrightarrow X \sqcup_\varphi Y$, and similarly for V as a subspace of $Y \hookrightarrow X \sqcup_\varphi Y$). Say such preimage is $W' \subseteq U$, then one has the following diagram:

$$\begin{tikzcd}
\mathcal{O}_U(W') & \mathcal{O}_Y(W \cap V) \arrow[l] \\
\mathcal{O}_X(W \cap U) \arrow[u] &
\end{tikzcd}$$

Hence, a natural choice for $\mathcal{O}_\varphi(W)$ is the fibre product of the above diagram:

$$\begin{tikzcd}
\mathcal{O}_U(W') & \mathcal{O}_Y(W \cap V) \arrow[l] & \\
\mathcal{O}_X(W \cap U) \arrow[u] & \mathcal{O}_\varphi(W) \arrow[l] \arrow[u] & \\
& & Z \arrow[ul, dotted, "\exists ! i"] \arrow[uul, bend right, "i_Y^*"'] \arrow[ull, bend left, "i_X^*"]
\end{tikzcd}$$

In particular, $\mathcal{O}_\varphi(W) = \{(s_1, s_2) : s_1|_{W \cap U} = s_2 \mid_{W \cap V}\}$ as product of rings.

Also, note that since $\mathcal{O}_\varphi(W)$ is constructed as the kernel of a product map, which is preserved under direct limits. Then, in particular the stalk of any poin $x \in U$, the induced stalk $\mathcal{O}_{\varphi,x}$ is the fibre product of $\mathcal{O}_{X,x} \to \mathcal{O}_{U,x}$ and $\mathcal{O}_{Y,x} \to \mathcal{O}_{U,x}$, which is also a local ring (and has local homomorphisms to $\mathcal{O}_{X,x}$ and $\mathcal{O}_{Y,x}$ respectively), based on the following lemma:

Lemma

Given A, B, C three local rings, and $\varphi : A \to C$, $\psi : B \to C$ two local homomorphisms, then the fibre product $A \times_C B$ with the two canonical projections $\pi_A : A \times_C B \to A$, $\pi_B : A \times_C B \to B$ are local ring, and local homomorphisms.

Proof: For simplicity, $A \times_C B := \{(a, b) \in A \times B \mid \varphi(a) = \psi(b)\}$ is the set theoretic fibre product.

Now, let $m_R \subseteq R$ ($R = A, B, C$) be the maximal ideal of R, we claim that the set $m_A \times_C m_B := (m_A \times m_B) \cap (A \times_C B)$ is the maximal ideal of $A \times_C B$:

- First, since the inclusion $A \times_C B \hookrightarrow A \times B$ is a ring homomorphism, with $m_A \times m_B$ being an ideal in $A \times B$, its preimage under the inclusion (which is $m_A \times_C m_B$) is automatically an ideal.

- Then, we claim that every element outside of $m_A \times_C m_B$ is a unit (which automatically makes $m_A \times_C m_B$ a maximal ideal). Take $(a, b) \in A \times_C B \setminus m_A \times_C m_B$. As set element one has $a \in A \setminus m_A$ and $b \in B \setminus m_B$. By the property of local rings, $a \in U(A)$ and $b \in U(B)$, so $a^{-1} \in A$ and $b^{-1} \in B$ exists.

  Now, notice the following equality:

$$\varphi(a^{-1}) = \varphi(a)^{-1} = \psi(b)^{-1} = \psi(b^{-1}) \tag{3.1}$$

  The middle equality is based on the fact that $(a, b) \in A \times_C B$ (so $\varphi(a) = \psi(b)$). Hence, one has $(a^{-1}, b^{-1}) \in A \times_C B$ serving as the inverse of (a, b), proving the claim.

Finally, about the local homomorphism part, notice that $\pi_A(m_A \times_C m_B) \subseteq m_A$ and $\pi_B(m_A \times_C m_B) \subseteq m_B$ set wise, automatically making them local homomorphisms. $\square$

This construction makes $\left(X \sqcup_\varphi Y, \mathcal{O}_\varphi\right)$ a locally ringed space.

Now, notice that for any point $a \in X \sqcup_\varphi Y$ (WLOG, say $a \in X$), pick an affine neighborhood $W \subseteq X$, and embedd it to $X \sqcup_\varphi Y$. Then, as a neighborhood one has that being a topological homeomorphism of the neighborhoods, which the fibre $\mathcal{O}_\varphi(W)$ turns out to also be isomorphic to $\mathcal{O}_X(W)$, proving that every point has an affine neighborhood.

Finally, the universal property is based on the fact that both the fibre product (in CRing) and adjoint space (in Top) has the corresponding universal property. And, the gathered morphism from $\mathcal{O}_Z \to \mathcal{O}_\varphi$ must be a morphism of presheaves, hence a morphism of sheaves.
$\square$

Okay, this last part is a bit shaky…maybe I need to come back at some point.

4 Projective Schemes